PHYSICS

1) In a certain system, the units of length, mass and time are chosen to be 1 cm, 100 g and 10 s. The unit of kinetic energy will be

(1) 0.1 erg
(2) 1 erg
(3) 10 erg
(4) None of these

2) A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β and comes to rest. If the total time elapsed is t, the maximum velocity acquired by the car will be :-

(1) $\frac{\alpha^2-\beta^2}{\alpha\beta}t$
(2) $\frac{\alpha^2+\beta^2}{\alpha\beta}t$
(3) $\frac{\alpha+\beta}{\alpha\beta}t$
(4) $\frac{\alpha\beta}{\alpha+\beta}t$

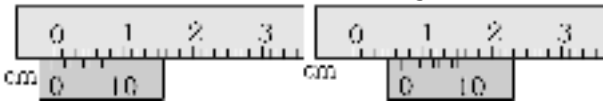
3) The period T of a soap bubble under SHM is given by $T = P^aD^bS^c$, where P is pressure, D is density and S is surface tension. Then the value of a, b and c are (pressure = force/area, surface tension = force/length) :-

(1) $-\frac{3}{2},\frac{1}{2},1$
(2) –1, –2, 3
(3) $\frac{1}{2},-\frac{3}{2},-1/2$
(4) $1,2,\frac{3}{2}$

4) In a particular system, the units of length, mass and time are chosen to be 10 cm, 10g and 0.1 s respectively. The value of 10N force in this system will be equivalent to :-

(1) 0.1 unit
(2) 1 unit
(3) 10 unit
(4) 100 unit

5) The main scale of a vernier callipers reads in millimeter and its vernier is divided into 10 divisions which coincides with 9 divisions of the main scale. The reading for shown situation is found to be (x/10) mm. Find the value of x.



(1) 65

(2) 69

(3) 80

(4) 75

6) Pathlength is always

(1) positive

(2) negative

(3) zero

(4) negative or positive

7) Given below are two statements :
Statement I : In a vernier callipers, one vernier scale division is always smaller than one main scale division.
Statement II : The vernier constant is given by one main scale division multiplied by the number of vernier scale division.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are false.

(2) Statement I is true but Statement II is false.

(3) Both Statement I and Statement II are true.

(4) Statement I is false but Statement II is true.

8) If $K = \frac{d^4x}{dt^4}$ find dimensional formula of K :-

(1) $[LT^{-3}]$

(2) $[L^4T^{-4}]$

(3) $[LT^{-4}]$

(4) [LT]

9) The order of magnitude of the number 1013 is:-

(1) 1

(2) 2

(3) 3

(4) 4

10) The resultant of $\vec{P}$ and $\vec{Q}$ is R_1. On reversing $\vec{Q}$, the resultant becomes R_2. What is value of $\left(R_1^2 + R_2^2\right)$?

(1) $P^2 + Q^2$

(2) $P^2 - Q^2$

(3) $2(P^2 + Q^2)$

(4) $2(P^2 - Q^2)$

11)

Select the incorrect statement :

(1) In unit conversion dimensional formula on both sides is same.

(2) A dimensionless quantity may have a unit.

(3) $v = \frac{x}{t}\sin\omega t$ can be derived by dimensional method.

(4) Value of dimensionless constant can not be found while deriving a relation by dimensional method.

12) A dimensionless quantity –

(1) never has a unit

(2) always has a unit

(3) may have a unit

(4) does not exist

13) Which of the following combination of three dimensionally different physical quantities P,Q,R can never be a meaningful quantity ?

(1) PQ – R

(2) $\frac{PQ}{R}$

(3) $\frac{P-Q}{R}$

(4) $\frac{PR-Q^2}{QR}$

14) For the pitch of a screw 0.1 cm and 200 divisions on circular scale. The least count will be :-

(1) 0.5 mm

(2) 0.05 mm

(3) 0.005 mm

(4) 0.0005 mm

15)

If area (A), velocity (v), and density (ρ) are base units, then the dimensional formula of force can be represented as

(1) $Av\rho$

(2) $Av^2\rho$

(3) $Av\rho^2$

(4) $A^2v\rho$

16) In a book, the answer for a particular question is expressed as b = $\frac{ma}{k}\left[\sqrt{1+\frac{2k\ell}{ma}}\right]$ here m represents mass, a represents acceleration, ℓ represents length. The unit of b should be

(1) m/s

(2) m/s^2

(3) meter

(4) /sec

17) If the energy (E),velocity (u) and force (F) be taken as the fundamental quantity, then the dimensions of mass will be :

(1) Fu^{-2}

(2) Fu^{-1}

(3) Eu^{-2}

(4) Eu^2

18) Which of the following conversion is true :-

(1) $1\ Kgm^2s^{-2} = 10^5\ gcm^2\ s^{-2}$

(2) 3×10^8 m/s = 3,00,000 km/h

(3) $6.67 \times 10^{-11}\ \frac{Nm^2}{kg^2} = 6.67 \times 10^{-8}\ \frac{dyne\ cm^2}{g^2}$

(4) $18\ \frac{km}{h^2} = 5\ m/s^2$

19) What is the dimensions of a and b in the relation F = $a\sqrt{x} + bt^2$, where F is a force, x is distance and t is time:

(1) $ML^{1/2}\ T^{-2}$, MLT^{-4}

(2) $ML^{1/2}T^{-1}$, MLT^{-4}

(3) $ML^{1/2}T^{-2}$, ML^2T^{-4}

(4) Both are dimensionless

20) Given that the displacement of an oscillating particle is given by y = Asin(Bx + Ct + D). The dimensional formula for ABCD is :-

(1) $[M^0L^{-1}T^0]$

(2) $[M^0L^0T^{-1}]$

(3) $[M^0L^{-1}T^{-1}]$

(4) $[M^0L^0T^0]$

21)

Velocity gradient $\frac{dV}{d\ell}$ in a liquid flow is related with the viscosity coefficient η, viscous force F and the surface area A of adjacent layers : $\frac{dV}{d\ell} = \frac{F}{\eta A}$. The dimensions of η are :

(1) $[MLT^{-1}]$

(2) $[ML^{-1}T^{-1}]$

(3) $[MLT^{-2}]$

(4) $[ML^{-1}T^{-2}]$

22) If E, M, L and G denotes energy, mass, angular momentum and universal gravitational constant, respectively, then EL^2/M^5G^2 represents the unit of :-

(1) Length

(2) Mass

(3) Time

(4) Angle

23)

If the energy, $E = G^ph^qc^r$, where G is the universal gravitational constant, h is the planck's constant c is the velocity of light, then the values of p, q and r are respectively.

(1) -1/2, 1/2 and 5/2

(2) 1/2, -1/2 and -5/2

(3) -1/2, 1/2 and 3/2

(4) 1/2, -1/2 and -3/2

24) The initial velocity of a particle is u and the acceleration is given by (kt), where k is a positive constant. The distance travelled in time t is :-

(1) $s = ut^2 + kt^2$

(2) $s = ut + \left(\frac{kt^3}{6}\right)$

(3) $s = ut + \left(\frac{kt^3}{2}\right)$

(4) $s = \left(\frac{ut^2}{2}\right) + \left(\frac{kt^3}{6}\right)$

25) If x = a – b, the maximum percentage error in the measurement of x will be :-

(1) $\left[\frac{\Delta a}{a} + \frac{\Delta b}{b}\right] \times 100\%$

(2) $\left[\frac{\Delta a}{a} - \frac{\Delta b}{b}\right] \times 100\%$

(3) $\left[\frac{\Delta a}{a-b} + \frac{\Delta b}{a-b}\right] \times 100\%$

(4) $\left[\frac{\Delta a}{a-b} - \frac{\Delta b}{a-b}\right] \times 100\%$

26) The mass and volume of a body are 4.237g and 2.5 cm^3 respectively. The density of the material of the body in correct significant figure is :-

(1) 1.6048 g cm^{-3}

(2) 1.69 g cm^{-3}

(3) 1.695 g cm^{-3}

(4) 1.7 g cm^{-3}

27) A body travels uniformly a distance of (24 ± 0.2)m in a time interval (6 ± 0.1)s. Then velocity of body is given by -

(1) (4 ± 0.2) m/s

(2) (4 ± 0.3) m/s

(3) (4 ± 0.05) m/s

(4) (4 ± 0.1) m/s

28) In an experiment the angles are required to be measured using an instrument, 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half-a degree (= 0.5°) then the least count of the instrument is :-

(1) half minute

(2) one degree

(3) half degree

(4) one minute

29) 4.338 + 4.835 × 3.88 ÷ 3.0 is equal to :

(1) 10.6

(2) 10.59

(3) 10.5912

(4) 10.591267

30) A dust particle oscillates in air with a time period which depends on atmospheric pressure P density of air d and energy of the particle E, then time period is proportional to

(1) $P^{\frac{5}{6}}d^{\frac{1}{2}}E^{\frac{1}{3}}$

(2) $P^{\frac{1}{2}}d^{3}E^{-2}$

(3) $P^{\frac{1}{3}}d^{\frac{1}{2}}E^{2}$

(4) $P^{2}d^{\frac{1}{2}}E^{-3}$

31) Which of the following sets cannot enter into the list of fundamental quantities in any newly proposed system of units?
(i) length, mass and velocity
(ii) pressure, density and velocity
(iii) force, velocity and time
(iv) force, momentum and time

(1) (i) and (iii)

(2) (ii) and (iv)

(3) only (iv)

(4) only (i)

32) The refractive index of water measured by the relation $\mu = \frac{\text{real depth}}{\text{apparent depth}}$ is found to have values of 1.34, 1.38, 1.32 and 1.36; the mean value of refractive index with percentage error :-

(1) 1.35 ± 1.48%

(2) 1.35 ± 0%

(3) 1.36 ± 6%

(4) 1.36 ± 0%

33) In the equation $\alpha = \frac{Fv^2}{\beta^2}\log_e\left[\frac{2\pi\beta}{v^2}\right]$.
F and v are force and velocity respectively. Dimensions of α and β are :-

(1) $[\alpha] = MLT^{+1}, \quad [\beta] = L^{-2}T^{+2}$

(2) $[\alpha] = ML^{-1}T^{0}, \quad [\beta] = L^{2}T^{-2}$

(3) $[\alpha] = MLT^{0}, \quad [\beta] = L^{2}T^{-2}$

(4) $[\alpha] = ML^{-1}T^{+1}, \quad [\beta] = L^{-2}T^{+2}$

34) In a system of units, unit of mass is 100 gm, unit of length is 10 cm unit of time is 0.5 sec then the value of unit of energy in this system, In terms of Joule. :-

(1) 250 Joule

(2) 25 Joule

(3) 0.004 Joule

(4) 0.04 Joule

35) Two resistance are measured in Ohm and is given as :-
$R_1 = 3\Omega \pm 1\%$ & $R_2 = 6\Omega \pm 2\%$
When they are connected in parallel, the percentage error in equivalent resistance is :-

(1) 3%

(2) 4.5%

(3) 0.67%

(4) 1.33%

36) The mass and volume of a body are found to be (5 ± 0.05) kg and (1 ± 0.05) m^3 respectively, then the maximum possible percentage error in density :-

(1) 3%

(2) 10%

(3) 5%

(4) 6%

37) The dimensional formula for acceleration, velocity and length are $\alpha\beta^{-2}$, $\alpha\beta^{-1}$ and $\alpha\gamma$. What is the dimensional formula of coefficient of friction :-

(1) $\alpha\beta\gamma$

(2) $\alpha^0\beta^{-1}\gamma^0$

(3) $\alpha^{-1}\beta^0\gamma^0$

(4) $\alpha^0\beta^0\gamma^{-1}$

38) If the units of length, velocity and force are halved, then the unit of power will be :-

(1) doubled

(2) halved

(3) one fourth

(4) remain unaffected

39) Two full turns of the circular scale of screw gauge cover a distance of 1 mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of –0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the 35th division of circular scale coincides with the reference line of main scale. The diameter of the wire is :

(1) 3.38 mm

(2) 3.32 mm

(3) 3.73 mm

(4) 3.67 mm

40) If L.C. of a vernier calliper is 0.2 mm. If main scale reading is 12 mm and 3rd division of vernier scale coincides with base line, find measured value by vernier calliper :

(1) 1.23 cm
(2) 1.26 cm
(3) 12.6 cm
(4) 12.3 cm

41) Significant figures in 75600 are-

(1) 2
(2) 3
(3) 6
(4) 7

42) A body travels uniformly a distance of (13.8±0.2) m in a time (4.0±0.3) s. The velocity of the body within error limits is :-

(1) (3.45 ± 0.2)m s^{-1}
(2) (3.45 ± 0.3)m s^{-1}
(3) (3.45 ± 0.4)m s^{-1}
(4) (3.45 ± 0.5)m s^{-1}

43) In a vernier callipers, one main scale division is x cm and n divisions of the vernier scale coincide with (n – 1) divisions of the main scale. The least count (in cm) of the callipers is :-

(1) $\left(\frac{n-1}{n}\right)x$
(2) $\frac{nx}{(n-1)}$
(3) $\frac{x}{n}$
(4) $\frac{x}{(n-1)}$

44) **Assertion (A) :** In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.
Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

45) The position x of a particle at time t is given by $x = \frac{V_0}{a}\left(1 - e^{at}\right)$, where V_0 is constant and $a > 0$. The dimensions of V_0 and a are

(1) M^0LT^{-1} and T^{-1}
(2) M^0LT^0 and T^{-1}
(3) M^0LT^{-1} and LT^{-2}
(4) M^0LT^{-1} and T

CHEMISTRY

1)

The volume of O_2 liberated at STP on decomposition of 12.25 g of $KClO_3$ completely

(1) 3.36 L
(2) 9.92 L
(3) 6.72 L
(4) 2.24 L

2)

Which of the following compounds illustrates the law of multiple proportions ?

(1) CO and CO_2
(2) NaCl and NaBr
(3) NaOH and KOH
(4) $ZnCO_3$ and $PbCO_3$

3) The Atomic weight of an element is 15. Its chloride has a vapour density of 60. The valency of the element is :-

(1) 3
(2) 5
(3) 7
(4) 9

4)

The number of moles of a gas in 1 m^3 of volume at NTP is :

(1) 4.46
(2) 0.446
(3) 1.464
(4) 44.6

5)

If 1 ml of water contains 20 drops. Then number of molecules in a drop of water is :

(1) 6.023×10^{23}

(2) 1.673×10^{21}

(3) 1.344×10^{18}

(4) 20 g

6) The empirical formula of a gaseous organic compound containing carbon and hydrogen is CH_2. The mass of one litre of this compound is exactly equal to that of one litre of N_2 At NTP. Therefore, the molecular formula of the compound is–

(1) C_2H_4

(2) C_3H_6

(3) C_6H_{12}

(4) C_4H_8

7)

0.01 mole of iodoform (CHI_3) reacts with Ag to produce a gas whose volume at 273°C and 1 atm is
$2CHI_3 + 6Ag \rightarrow C_2H_2 + 6AgI(s)$

(1) 224 ml

(2) 112 ml

(3) 336 ml

(4) 448 ml

8)

The volume of gas at NTP produced by 100 gm of CaC_2 with water:-
$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$

(1) 70 litre

(2) 35 litre

(3) 17.5 litre

(4) 22.4 litre

9) Equivalent weight of a divalent metal is 24. The volume of hydrogen liberated at STP by 12 gms of the same metal when added to excess of an acid solution is

(1) 2.8 litres

(2) 5.6 litres

(3) 11.2 litres

(4) 22.4 litres

10)

12 lit. of H_2 and 11.2 lit. of Cl_2 are mixed and exploded. The composition by volume of mixture is–

(1) 24 lit. of HCl (g)

(2) 0.8 lit. Cl_2 and 20.8 lit. HCl (g)

(3) 0.8 lit. H_2 and 22.4 lit. HCl (g)

(4) 22.4 lit. HCl (g)

11)

Which of the following will not have a mass of 10 gm :-

(1) 0.1 mol of $CaCO_3$

(2) 1.5×10^{23} Ca^{2+} ions

(3) 0.016 mol of CO_3^{2-}

(4) 6.02×10^{22} molecules of $CaCO_3$

12)

No. of molecules of $Na_3(PO_4)_2$ can formed by using 0.4 mol of phosphorous atom :-

(1) 12×10^{23}

(2) 1.2×10^{23}

(3) 0.12×10^{23}

(4) 1.2×10^{24}

13)

Which one the following has heighest no. of oxygen atoms :-

(1) 0.1 mol Fe_2O_3

(2) 0.2×10^{23} molecule of $K_2Cr_2O_7$

(3) 0.5 g H_2O

(4) All have equal no. of oxygen atoms

14) An organic compound contains 28.5% Na, 40% S, 31.5% O and its vapour density is 79. What is its molecular formula ?

(1) NaS_2O_2

(2) $Na_2S_2O_2$

(3) $Na_2S_2O_3$

(4) $Na_2S_3O_2$

15) If 6.3 g of $NaHCO_3$ is added to 10.0 g of HCl solution, the residue is found to weight 12.0 g. What is the weight of CO_2 released in this reaction ?

(1) 44 g

(2) 22 g

(3) 11 g

(4) 4.3 g

16)

H_2SO_4 can be prepared in following two step process : [% yield of each step is indicated]

(A) $S(g) + O_2(g) \xrightarrow{50\%\ yield} SO_2(g)$

(B) $NO_2(g) + SO_2(g) + H_2O(\square) \xrightarrow{80\%\ yield} H_2SO_4(aq.) + NO(g)$

What mass of sulphur is required to prepare 1960 gm H_2SO_4 ?

(1) 640 gm

(2) 1450 gm

(3) 1600 gm

(4) 1280 gm

17) A gaseous mixture contains $CO_2(g)$ and $N_2O(g)$ in a 2 : 5 ratio by mass. The ratio of the number of molecules of $CO_2(g)$ and $N_2O(g)$ is :

(1) 5 : 2

(2) 2 : 5

(3) 1 : 2

(4) 5 : 4

18) Two oxides of a metal contains 50% and 40% of the metal respectively. The formula of the first oxide is MO. Then the formula of the second oxide is

(1) MO_2

(2) M_2O_3

(3) M_2O

(4) M_2O_5

19)

A certain compound has the molecular formula X_4O_6. If 10g of X_4O_6 has 5.72 g of X, atomic mass of X is :-

(1) 32 amu

(2) 37 amu

(3) 42 amu

(4) 98 amu

20) On analysis, certain compound was found to contain 254g of (P) and 80g of (Q). If the atomic weight of (P) is 127 and that of (Q) is 16, then formula of the compound containing (P) and (Q) is :

(1) P_2Q_5

(2) P_5Q_2

(3) P_2Q

(4) PQ

21) Percentage loss in mass on heating mixture of Na_2CO_3 & $CaCO_3$ containing equal mass of the two components will be

(1) 44%
(2) 22%
(3) 35%
(4) 50%

22)

Which of the following has maximum mass –

(1) 0.1 g atom of C
(2) 0.1 mole of NH_3
(3) 6.02×10^{22} molecule of H_2 gas
(4) 1120 ml of CO_2 at 1 atm, 273K

23) At room temperature, the density of water is 1 g/ml and density of ethanol is 0.8 g/ml. What volume of ethanol contains the same number of molecules as are present in 175 ml of water.

(1) $\simeq$ 565 ml
(2) $\simeq$ 460 ml
(3) 360 ml
(4) $\simeq$ 742 ml

24) 1 L of a gas at STP has 1.25 g mass. The gas can be :-

(1) N_2
(2) C_2H_2
(3) O_2
(4) CH_4

25) A compound contains 38.8% C, 16.0% H and 45.2% N by mass. The empirical formula of the compound would be :-

(1) CH_3NH_2
(2) CH_3CN
(3) C_2H_5CN
(4) $CH_2(NH)_2$

26)

1.0 g of hydrogen contains 6×10^{23} atoms. The atomic weight of helium is 4. It follows that the number of atoms in 1 g of He is :-

(1) $1/4 \times 6 \times 10^{23}$

(2) $4 \times 6 \times 10^{23}$

(3) 6×10^{23}

(4) 12×10^{23}

27)

Atom of oxygen in Fe_2O_3 and FeO is in the simple ratio of :-

(1) 3 : 2

(2) 1 : 2

(3) 2 : 1

(4) 3 : 1

28) The vapour density of chloride of element is 39.5. The equivalent weight of the elements is 3.82. The atomic weight of the element is :-

(1) 15.28

(2) 7.64

(3) 3.82

(4) 11.46

29) A polystyrene having formula $Br_3C_6H_3(C_3H_8)_n$ found to contain 10.46% of bromine by weight. The value of n :- (at. wt of Br = 80)

(1) 40

(2) 43

(3) 45

(4) 47

30) Elements A and B form two compounds B_2A_3 and B_2A. 0.05 moles of B_2A_3 weight 9.0 gms and 0.10 mole of B_2A weight 10 gms atomic weight of A and B are-

(1) 20 and 30

(2) 30 and 40

(3) 40 and 30

(4) 30 and 20

31) Number of moles of water in 488 gm of $BaCl_2.2H_2O$ are : (Ba = 137)

(1) 2 moles

(2) 4 moles

(3) 3 moles

(4) 5 moles

32) **Assertion :** CO_2 has more number of molecule in its 88 g than number of molecule of H_2O in its

72 g.
Reason : Both have 4 mole of molecules.

(1) Both Assertion and Reason are correct.
(2) Assertion is correct but Reason is incorrect.
(3) Assertion is incorrect but Reason is correct.
(4) Both Assertion and Reason are incorrect.

33) **Assertion :** Specific gravity is dimensionless.
Reason : Specific gravity is density of a substance measured w.r.t. density of water at 4°C.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

34) **Assertion :** A compound contains 3.2% oxygen by mass, then minimum molecular mass of the compound is 500.
Reason : For minimum molecular mass of the compound, it must have one atom of that element.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

35) **Assertion :** If we have equal mass of each CH_4, O_2 and SO_2 in a container, then number of mol of CH_4, O_2 and SO_2 are in 4 : 2 : 1 mole ratio.
Reason : Molecular mass of CH_4, O_2 and SO_2 are in 1 : 2 : 4 ratio.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

36) **Assertion :** Average atomic mass of B^{10} and B^{11} is always between 10 and 11.
Reason : Greater the precentage of B^{10}, more will be the average atomic mass.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

37) **Assertion :** 1 gm molecules of CH_4 contains 5 N_A atoms.
Reason : 1 mole of CH_4 weighs 16 g.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

38) **Assertion :** Molarity of pure water is 55.55 M at 298 K.
Reason : Molarity is temperature dependent.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

39) **Assertion :** $H_3PO_4 + NaOH \rightarrow NaH_2PO_4 + H_2O$ in this reaction the equivalent weight of H_3PO_4 is M/1.
Reason : H_3PO_4 is tribasic acid.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

40) In the reaction,
Assertion (A): The volume of NaOH solution added to 12 g of $NaH_2\ PO_4$ to convert it into Na_3PO_4 is 200 ml
Reason (R) : Number of H^+ replaced in above reaction is 2.

(1) A and R both are correct and R is the correct explanation of A

(2) A and R both are correct but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

41) How many moles of electron weight one kilogram?

(1) 6.02×10^{23}

(2) $\frac{10^{31}}{9.1}$

(3) $\frac{6.02 \times 10^{54}}{9.1}$

(4) $\frac{10^{8}}{9.1 \times 6.02}$

42) 20 mL of a gaseous hydrocarbon was exploded with 120 mL of oxygen. A contraction of 60 mL was observed, and a further contraction of 60 mL took place when KOH was added. What is the formula of the hydrocarbon :

(1) C_3H_6

(2) C_3H_8

(3) C_2H_6

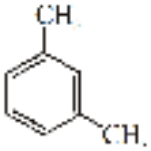
(4) C_4H_{10}

43) Equivalent weights of X_2Y and X_2Y_3 are 38 and 18 respectively. Find the atomic masses of X and Y :-

(1) 30, 8
(2) 30, 16
(3) 10, 16
(4) 16, 10

44) 2.0 g sample contain mixture of SiO_2 and Fe_2O_3, on very strong heating leave a residue weighing 1.96 g. The reaction responsible for loss of weight is $Fe_2O_3(s) \rightarrow Fe_3O_4(s) + O_2(g)$, What is the percentage by mass of SiO_2 in original sample?
[Fe = 56, Si = 28]

(1) 20%
(2) 60%
(3) 80%
(4) 40%

45) (2.12 gm) of  (meta-xylene) on complete combustion gives $CO_2(g)$ and $H_2O(g)$. Find ratio of number of atoms O to C in product formed.

(1) $\frac{1}{2}$
(2) $\frac{21}{8}$
(3) $\frac{19}{10}$
(4) $\frac{9}{2}$

BIOLOGY

1) Phospholipids are arranged with in the cell membrane with the :-

(1) Polar head towards the inner sides and tail towards the outer sides
(2) Polar head towards the inner sides and tail towards the inner sides
(3) Polar head towards the outer sides and tail towards the inner sides
(4) Polar head towards the outer sides and tail towards the outer sides

2) Algae have cell wall made up of:

(1) Cellulose, hemicellulose and pactins

(2) Cellulose, galactans and mannans and $CaCO_3$

(3) Hemicellulose, pectins and proteins

(4) Pectins, cellulose and proteins

3) **Assertion:** Lipids are arranged within the cell membrane with the hydrophobic tails towards the inner part.
Reason: This ensures that non-polar tail of saturated hydrocarbons is protected from the aqueous environment.

(1) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are correct but Reason is incorrect explanation of Assertion.

(3) Assertion is correct but reason is incorrect

(4) Both assertion and reason are incorrect

4) **Assertion:** Active transport requires the expenditure of energy by the cell.
Reason: Active transport is the movement of substances across the plasma membrane against the concentration tradient.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

5) Which one of the following structures between two adjacent cells is an effective transport pathway?

(1) Plasmalemma

(2) Plasmodesmata

(3) Plastoquinones

(4) Endoplasmic reticulum

6) Cell wall in plant is
A. Permeable
B. Living
C. Mainly made up of lipo protein
D. Mainly made up of polysaccharide

(1) A, B, C & D

(2) A, B & D

(3) A & D

(4) A, C & D

7) Ribophorin helps in attachment of ribosome to E.R. It is a

(1) Glycoprotein

(2) Glycolipid

(3) Lipid

(4) Carbohydrate

8) How many statements are correct ?
(A) Animal cells contain membrane bound organelle called centriole which help in cell division.
(B) Lipids are arranged within the plasma membrane with the polar head towards the inner sides and the hydrophobic tails towards the outer side.
(C) The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighboring plant cells.
(D) The quasifluid nature of lipid enables lateral movement of proteins within the overall bilayer.

(1) One

(2) Two

(3) Three

(4) Four

9) The stroma of chloroplast contains:-
(A) Enzyme for carbohydrates and proteins synthesis.
(B) Small, single stranded DNA molecule
(C) Ribosomes of 70's type

(1) Only A

(2) Only B

(3) A and C only

(4) All of the above

10) Match the **Column-I** and **Column-II** and choose the **correct** option :

Column-I (Plastid)		**Column-II (Stored nutrient or pigment)**	
(a)	Amyloplast	(i)	Chlorophyll
(b)	Chromoplast	(ii)	Proteins
(c)	Aleuroplast	(iii)	Carbohydrate
(d)	Chloroplast	(iv)	Carotenoids

(1) a-iii, b-i, c-ii, d-iv

(2) a-ii, b-i, c-iii, d-iv

(3) a-ii, b-iv, c-iii, d-i

(4) a-iii, b-iv, c-ii, d-i

11) (a) ER divide the intracellular space into two distinct compartments i.e. luminal and extra luminal.
(b) SER is the major site for synthesis of protein
(c) Golgi bodies are consist of cisternae which have diameter between 0.5-1 μm.
Choose the **correct** statements -

(1) a, b, c

(2) a, b

(3) b, c

(4) a, c

12) Study the following statements regarding mitochondria and select the **correct** ones :
(i) These are the sites of aerobic respiration.
(ii) Matrix contains single circular ds DNA molecule, a few RNA molecules, 70S ribosomes.
(iii) Mitochondria divide by fission.
(iv) Mitochondria are fully-autonomous.

(1) (i) and (ii)

(2) (iii) and (iv)

(3) (i), (ii) and (iii)

(4) All of the above

13) As the polar molecules can not pass through the nonpolar ....."I"....., they requires ....."II".... of the membrane to facilitate their transport across the membrane:-
Choose the correct word for I and II :-

	I	**II**
(1)	Cholesterol	Carbohydrate
(2)	Protein	Lipid
(3)	Carbohydrate	Protein
(4)	Lipid bilayer	Protein

(1) 1

(2) 2

(3) 3

(4) 4

14)

Identify structure A, B and C in given diagram:-



(1) A-SER, B-Cytosol, C-RER

(2) A-SER, B-Nuclear pore, C-RER

(3) A-RER, B-Nuclear pore, C-SER

(4) A-RER, B-Ribosome, C-SER

15) How many of the given features are associated with both plastid and mitochondria ?

Thylakoids, Cristae, Double membrane,
70 S ribosomes, RuBisCO, ss DNA, Stroma

(1) Two
(2) Three
(3) Four
(4) Five

16) Which of the following is correct regarding percentage of proteins, lipids and carbohydrates in plasma membrane ?

(1) Proteins < Lipids < Carbohydrates
(2) Lipids < Proteins < Carbohydrates
(3) Carbohydrates < Proteins > Lipids
(4) Lipids < Carbohydrates < Proteins

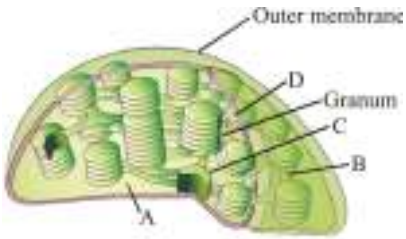
17)

Where are porins are present :-

(1) Outer membrane of plastid
(2) Inner membrane of mitochondria
(3) Inner membrane of plastids
(4) Outer membrane of ribosomes

18) Middle lamella is

(1) Present inside the primary wall
(2) Present inside the secondary wall
(3) Present outside the primary wall
(4) Present in between secondary and tertiary walls



19)
Above diagram showing the sectional view of chloroplast. Chlorophyll pigments and enzymes for the synthesis of carbohydrates are found respectively in :-

(1) A and B parts of diagram
(2) D and B parts of diagram
(3) C and A parts of diagram
(4) B and C parts of diagram

20) Fluid nature of membrane is able to explain

(1) cell growth, cell division
(2) formation of intercellular junctions
(3) secretion and endocytosis
(4) All of the above

21) Part of endomembrane system is :

(1) Chloroplast, Mitochondria
(2) ER, Golgi body, Microbodies
(3) ER, Golgi body, Lysosome, Vacuole
(4) ER, Nucleus, PM

22) Choose the incorrect statement :-

(1) Primary cell wall is first formed cell wall
(2) Secondary cell wall is thick and rigid layer
(3) Tertiary cell wall is present in all plant cells
(4) Middle lamella is the common layer between two adjacent plant cells

23)

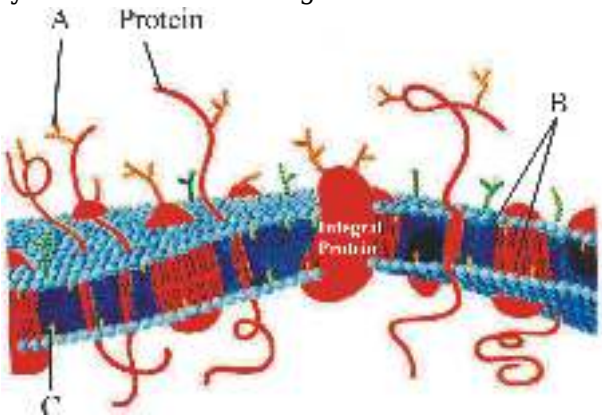
Match the Column–I with Column–II.

	Column–I		Column–II
(A)	Lysosomes	(i)	Photophosphorylation
(B)	Mitochondria	(ii)	Protein synthesis
(C)	Chloroplast	(iii)	Hydrolytic enzymes
(D)	Ribosomes	(iv)	Oxidative phosphorylation

Options :

(1) (A)–(iii), (B)–(i), (C)–(ii), (D)–(iv)
(2) (A)–(iii), (B)–(iv), (C)–(i), (D)–(ii)
(3) (A)–(ii), (B)–(iv), (C)–(i), (D)–(iii)
(4) (A)–(i), (B)–(iii), (C)–(iv), (D)–(ii)

24) Identify the correct matching for functions related to the cell membrane components marked as A, B & C.



	A	B	C

(1)	Cell to cell recognition	Membrane fluidity	Stability
(2)	Stability	Maintains membrane stability & fluidity at different temp.	Flip-flop movement
(3)	Cell to cell recognition	Maintains membrane stability & fluidity at different temp.	Channel
(4)	Stability	Membrane stability	Transport

(1) 1

(2) 2

(3) 3

(4) 4

25) Cell wall not only gives _____ to the cell. It also help in _____ and provide barrier to undesirable _____.
Option :
(i) Macromolecules
(ii) Cell to cell recognition
(iii) Micromolecules
(iv) Shape
(v) Cell to cell interaction
The correct sequence is :-

(1) (iv), (iii), (i)

(2) (iv), (v), (i)

(3) (iv), (iii), (ii)

(4) (iii), (ii), (i)

26) The stroma of chloroplast have :

(1) Ribosome

(2) DNA

(3) Enzymes for dark reaction

(4) All of these

27) Active transport involves

(1) Production of ATP

(2) Requirement of energy

(3) Production of toxin

(4) Release of energy

28) Keeping in view the "fluid mosaic model" for the structure of cell membrane, which one of the following statements is ***correct*** with respect to the movement of lipids and proteins from one lipid monolayer to the other (described as flipflop movement) ?

(1) While proteins can flip-flop, lipids can not

(2) Neither lipids, nor proteins can flip-flop

(3) Both lipids and proteins can flip-flop

(4) While lipids can rarely flip–flop,proteins can not

29) F_1 particle found in

(1) Inner membrane of chloroplast and thylacoid membrane.

(2) Outer membrane of mitochondria and in matrix.

(3) In inner membrane of mitochondria toward matrix side.

(4) In mitochondrial inner membrane toward perimitochondrial space side (PMS).

30) A cell organelle 'X' is divided into two types on the basis of a cell organelle 'Y', that helps in the protein synthesis. Identify 'X' and 'Y' respectively:

(1) Golgi complex and ribosome

(2) ER and mitochondria

(3) ER and ribosome

(4) Lysosome and ER

31) Depending upon the ..... , membrane proteins can be classified as integral or peripheral :-

(1) Size

(2) Sedimentation rate

(3) Ease of extraction

(4) Molecular weight

32) In human beings, percentage of lipid and protein in membrane of erythrocyte :

(1) 40% and 58% respectively

(2) 42% and 59% respectively

(3) 40% and 52% respectively

(4) 52% and 40% respectively

33) Red colour of tomato is due to are:

(1) Chl a

(2) Xanthophylls

(3) Chl b

(4) Lycopene

34) Match the column-I with column-II :-

	Column-I		**Column-II**
a	Golgi apparatus	i	Synthesis of protein
b	Lysosomes	ii	Trap waste and excretory products

c	Vacuoles	iii	Formation of glycoproteins and glycolipids
d	Ribosomes	iv	Digesting biomolecules

Choose the right match from options given below :-

(1) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(4) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

35) Mitochondria :-
(a) are easily visible under the microscope
(without specifically stained)
(b) are typically sausage-shaped or cylindrical
(c) are double membrane bound structures
(d) have two aqueous compartments

(1) a, d correct and b, c incorrect

(2) a, b correct and c,d incorrect

(3) a incorrect and b, c, d correct

(4) a, d incorrect and b, c correct

36) **Assertion:** Mitochondria and chloroplasts are semi-autonomous organelles.
Reason: Mitochondria and chloroplast are devoid of ribosomes.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

37) **Assertion:** In animal cells, the lipid-like steroidal hormones are synthesized in SER.
Reason: The smooth ER is the major site for synthesis of lipid.

(1) If both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) If both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) If the assertion is true but the reason is false.

(4) If both the assertion and reason are false.

38) **Assertion :** It is believed that mitochondria shows endosymbiotic origin with bacteria.
Reason : Mitochondria contains 70S ribosomes and divides by binary fission.

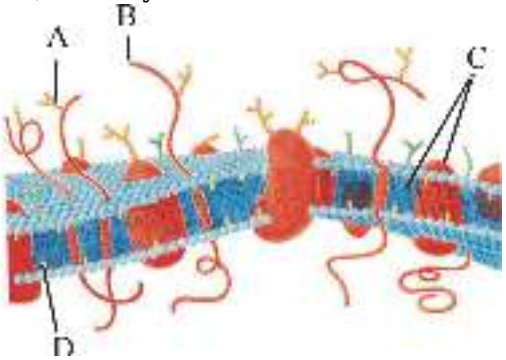
(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

39) Identify the A, B, C and D select the correct statements regarding to A, B, C and D :-



(1) Percentage of "A" is maximum in the membrane

(2) Percentage of "B" is minimum in the membrane

(3) Lateral movement of "A" help in the measurement of fluidity of membrane

(4) The fluidity of membrane is due to "C"

40) **Statement-I :** Neutral solutes may move, across the membrane by the process of simple diffusion along the concentration gradient i.e. from higher concentration to the lower.
Statement-II : The polar molecules can not pass through the polar lipid bilayer, they require a carrier protein of the membrane to facilitate their transport across the membranes.

(1) Statement I and II both are correct

(2) Statement I and II both are incorrect

(3) Only Statement I is correct

(4) Only Statement II is correct

41) **Statement I :** Plastids bear some specific pigments, thus imparting specific colours to the plants.
Statement II : The leucoplasts are the colourless plastids of varied shapes and sizes with stored nutrients.

(1) Statement (I) and (II) both are correct.

(2) Statement (I) and (II) both are incorrect.

(3) Statement (I) is correct and (II) is incorrect.

(4) Statement (II) is correct and (I) is incorrect.

42) **Statement (I) :** A number of organised flattened membranous sacs called the thylakoids, are present in stroma.
Statement (II) : The membrane of the thylakoids enclose a space called lumen.

(1) Statement (I) and statement (II) both are correct.

(2) Statement (I) and statement (II) both are incorrect.

(3) Statement (I) is correct and (II) is incorrect.

(4) Statement (II) is correct and (I) is incorrect.

43) **Statement-I :** Plastids are found in all plant cells and in Euglenoides.

Statement-II : Majority of the chloroplasts of the green plants are found in the mesophyll cells of the leaves.

(1) Statement-I and statement-II both are true
(2) Statement-I is true and statement-II is false
(3) Statement-I is false and statement-II is true
(4) Statement-I and statement-II both are false

44) The convex-face of cisternae of golgi body is also known as:-

(1) Cis-face
(2) Forming face
(3) Trans-face
(4) Both 1 and 2

45) What is the main function of cristae?

(1) To hold the vesicles formed
(2) Increase the surface area
(3) Increase the density of organelle
(4) All of these

46) Consider the following statements regarding white adipose tissue :-
(a) It is characterized by a single large droplet of fat in the cytoplasm of its cells
(b) Cells contain many mitochondria
(c) It is found in new born baby and hibernating mammals
(d) Cytoplasm is pushed to periphery
Identify correct statement :

(1) a and d
(2) a and b
(3) b and c
(4) b, c and d

47) The kind of tissue that forms the supportive structure in our pinna (external ears) is also found in :-

(1) Tip of the nose
(2) Vertebrae
(3) Nails
(4) Ear ossicles

48) Adjacent epithelial cells are held together by means of

(1) Lipsomes
(2) Glyoxisomes/Glyoxysomes
(3) Desmosomes

(4) Microsomes

49) Consider the following statements (i) – (iii) each with two blanks.
(i) Pseudostratified epithelium lines the (1) tract while transitional epithelium lines the (2) tract.
(ii) Lacunae of bones house (3) while lacunae of cartilage contain (4) .
(iii) Tendon contains bundles of (5) fibres and row of (6) cells between them.
Which one of the following options, gives the correct fill ups for the respective blank numbers from (1) to (6) in the statements?

(1) (1) – respiratory, (2) – urinary, (5) – white, (6) – fibroblast

(2) (1) – urinary, (2) – respiratory, (3) – osteocytes, (4) – chondrocytes

(3) (3) – chondrocytes, (4) – osteocytes, (5) – yellow, (6) – fibroblast

(4) (1) – respiratory, (2) – urinary, (3) – chondrocytes, (4) – osteocytes

50) Out of following how many are the secretions of exocrine glands :-
Mucus, Saliva, Adrenaline, Ear wax, Vasopressin, Oil, milk, pepsin

(1) Four

(2) Five

(3) Six

(4) Seven

51)

While taking viva voce at the zoology practicals, professor SM presented four different structures before the students on a tray labelled as :
(A) White Regular (B) Yellow Regular
(C) White Irregular (D) Yellow Irregular
He then asked his students to perform two different tasks.
(i) To pick up any of the four structures that is meant for connecting a bone to another bone.
(ii) To pick up any of the four structures that is meant for connecting a muscle to a bone.

Identify the correct option.

(1) (i) – B and (ii) – C

(2) (ii) – A and (i) – B

(3) (ii) – A and (i) – D

(4) (ii) – B and (i) – A

52) Which of the following is false about columnar epithelium :- (I) It's made up of tall and slender cells
(II) Free end surface may have microvilli
(III) They are found in stomach and intestine so helps in absorption, secretion and excretion
(IV) Ciliated epithelium is mainly present in hollow structure like fallopian tube
(V) They have apical nuclei

(1) Only V

(2) III, V

(3) II, III, V

(4) III, IV, V

53) Which of the following statements about cell junction are correct :-
(i) All the cells of the epithelium are held together with little intercellular materials.
(ii) In nearly all animal tissues specialized junction provide both structural and functional link between its individual cells.
(iii) Tight junction help to stop substances from leaking across a tissue.
(iv) Adhering junctions provide cementing to keep neighbouring cells together.
(v) Gap junctions provide cytoplamic channels between cells for passage of ions, small molecules and sometimes big molecules.

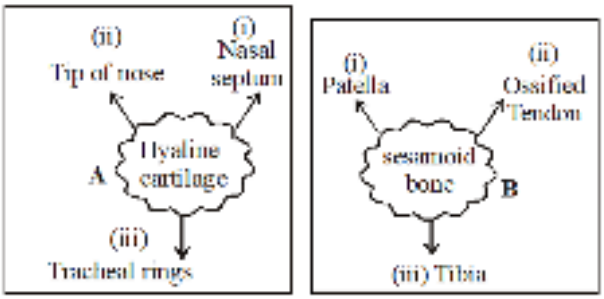
(1) (ii) and (iii)

(2) (i), (ii) and (iii)

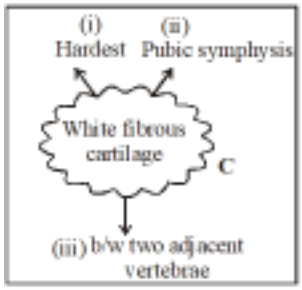
(3) (iv) and (v)

(4) (i), (ii), (iii), (iv) and (v)

54) Following A, B and C structures are surrounded by their suitable examples found in human body.



Which one is incorrect regarding to them :-



(1) A-(ii) B-(ii) C-(iii)

(2) A-(ii) B-(iii) C-(i)

(3) A-(i) B-(ii) C-(iii)

(4) A-(ii) B-(i) C-(iii)

55) Assertion : The infant does not shiver in cold temperature.
Reason : They have brown fat.

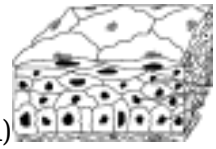
(1) If both Assertion & Reason are correct & Reason is correct explanation of Assertion

(2) If both Assertion & Reason are correct & Reason is not correct explanation of Assertion

(3) If Assertion is true & Reason is false

(4) If both Assertion & Reason is false

56) Observe the given diagrammatic combination and select correct option.

(A)  – Cover the dry surface of the skin and dry surface of buccal cavity.

(B)  – Secrete saliva, earwax and its example is goblet cells of the alimentary canal

(1) A and B both are correct

(2) A is correct and B is incorrect

(3) A and B both are incorrect

(4) A is incorrect and B is correct

57) Match the following column?

	Column-I		**Column-II**
(A)	Dense irregular connective tissue	(i)	Solid, Pliable and resist compression
(B)	Areolar loose connective tissue	(ii)	Fibroblasts and mostly collagen fibre oriented differently
(C)	Specialized connective tissue (cartilage)	(iii)	Blood
(D)	Fluid connective tissue	(iv)	Cells and fibre loosely arrange and present beneath the skin

(1) A–(i); B–(iv); C–(iii); D–(ii)

(2) A–(iv); B–(i); C–(iii); D–(ii)

(3) A–(ii); B–(iv); C–(i); D–(iii)

(4) A–(iii); B–(i); C–(iv); D–(ii)

58) Choose incorrect statement :
(a) All gland are multicellular glands except goblet cells and paneth cells.
(b) Epididymis have non-ciliated epithelium.
(c) Inner lining of duct of salivary gland is made up of more than one layer of cells.
(d) Tendon attach skeletal muscle to bone is a example of loose connective tissue.

(1) (a) and (b)

(2) (b) and (d)

(3) (c) and (d)

(4) (b), (c) and (d)

59) Match the list-I with list-II

	List-I		List-II
(a)	Dense regular connective tissue	(i)	In the skin
(b)	Specialised connective tissue	(ii)	Tendon
(c)	Dense Irregular connective tissue	(iii)	Bones
(d)	Loose connective tissue	(iv)	Ligament
		(v)	Beneath the skin
		(vi)	Blood

(1) (a)-(ii),(iii), (b)-(iv),(vi), (c)-(v), (d)-(i)

(2) (a)-(v),(ii), (b)-(iii),(vi), (c)-(i), (d)-(iv)

(3) (a)-(ii),(iv), (b)-(iii),(vi), (c)-(i), (d)-(v)

(4) (a)-(ii), (iv), (b)-(i),(iii), (c)-(v), (d)-(vi)

60) **Statement-I** : Neuroglia make up more than one half the volume of neural tissue
Statement-II : A sheath of tough connective tissue encloses several bundles of muscle fibers
Select the correct option –

(1) Statement-I is incorrect and statement-II is correct

(2) Statement-II is incorrect and statement-I is correct

(3) Both statement-I and statement-II are correct

(4) Both statement-I and statement-II are incorrect

61) Read the following statement about connective tissue.
(a) In all connective tissue except blood, the cells secretes fibers of structural proteins called collagen or elastin.
(b) Tendons and ligaments are the example of dense regular connective tissue
(c) Loose connective tissue has cells and fibres loosely arranged in a semi fluid ground substance
(d) Dense regular connective tissue is also found in skin.
Choose the set of correct statements.

(1) a and d

(2) b and d

(3) a, b and c

(4) b, c and d

62) **Assertion :-** Skeletal muscle tissue is closely attached to skeletal bones.
Reason :- Bone have a hard and non-pliable ground substances rich in calcium.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

63)

Match list I with list II and find the correct option:-

List-I (Epithelial tissue)		**List-II (Location)**	
A	Cuboidal	I	Epidermis of skin
B	Ciliated	II	Inner lining of blood vessels
C	Columnar	III	Inner surface of gall bladder
D	Squamous	IV	Inner lining of Fallopian tube
E	Keratinized squamous	V	Lining of pancreatic duct

(1) (A)-(V), (B)-(IV), (C)-(II), (D)-(III), (E)-(I)

(2) (A)-(III), (B)-(IV), (C)-(V), (D)-(II), (E)-(I)

(3) (A)-(V), (B)-(IV), (C)-(III), (D)-(II), (E)-(I)

(4) (A)-(III), (B)-(IV), (C)-(V), (D)-(I), (E)-(II)

64) Intercalated discs are characterstics of muscles that are found in :-

(1) Brain

(2) Spinal cord

(3) Stomach

(4) Heart

65) Which points are correct for smooth muscles ?
(A) They are branched
(B) They are unstriated
(C) They are involuntary
(D) Intercalated disc are present

(1) A, B

(2) B, C

(3) C, D

(4) A, D

66) Which of the following is not a feature of skeletal muscle ?

(1) Striations

(2) Voluntary control

(3) Non-fatigue nature

(4) All are features of skeletal muscle

67) Smooth muscle fibres are–

(1) Cylindrical, unbranched, striated, multinucleate and voluntary

(2) Spindle-shaped, unbranched, non-striated, uninucleate and involuntary

(3) Cylindrical, unbranched, non-striated, multinucleate and involuntary

(4) Spindle-shaped, unbranched, striated, uninucleate and voluntary

68) Which of the following properties of cardiac muscles are correct ?
(i) These are the muscles of the heart.
(ii) These are non-striated.
(iii) These are involuntary in their functions.
(iv) These are controlled by nervous system directly.
(v) Cardiac muscles are voluntary.
Select the correct option.

(1) (i) and (iii)

(2) (ii), (iv) and (v)

(3) (i), (iv) and (v)

(4) (ii) and (iii)

69) Which of the following statements are incorrect w.r.t. neuron?
(A) The proximal end of the axon is branched.
(B) Nissl's granules are present in the cell body as well as dendrites.
(C) The axon transmit nerve impulse towards the cell body.
(D Multipolar neurons are found in the cerebral cortex.

(1) A, B and D

(2) Only A and D

(3) B, C and D

(4) Only A and C

70) Given below are assertion and reason. Mark the correct choice from given options.
Assertion : The myelinated nerve fibres are enveloped with Schwann cells, which form a myelin sheath around the axon.
Reason : Unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon.

(1) Assertion and reason both are true and the reason is correct explanation of assertion

(2) Assertion and reason both are true but the reason is not correct explanation of assertion

(3) Assertion is true but reason is wrong

(4) Assertion and reason both are wrong

71) Match List-I with List-II

	List-I		List-II
(a)	Multipolar neuron	(i)	Somatic neural system
(b)	Bipolar neuron	(ii)	Cerebral cortex
(c)	Myelinated nerve fibre	(iii)	Retina of Eye
(d)	Unmyelinated nerve fibre	(iv)	Spinal nerves

Choose the **correct answer** from the options given below:

(1) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

(2) (a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

(3) (a) - (ii), (b) - (iii), (c) - (i), (d) - (iv)

(4) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

72) Nissl's granules are found in____ and helps in _____ :-

(1) Cyton only, Lipid synthesis

(2) Cyton & Axon, Protein synthesis

(3) Dendron & Axon, Protein synthesis

(4) Dendron & cyton, Protein synthesis

73) Schwann cells are found on :-

(1) Myelinated nerve fibres

(2) Unmyelinated nerve fibres

(3) Both (1) and (2)

(4) Cell body of neuron

74) Which of the characteristic is shown by neuron:

(1) Contractibility

(2) Elastiity

(3) Excitability

(4) Erectability

75) Match the following :-

(A)	Small phagocytic neuroglia	(i)	Schwann cells
(B)	Provide myelin sheath for CNS axons	(ii)	Satellite cells
(C)	A cluster of cell bodies in CNS	(iii)	Microglia
(D)	Provide myelin sheath for PNS axons	(iv)	Nuclei
		(v)	Oligodendrocytes
		(vi)	Ganglia

(1) A – (i), B – (ii), C – (iii), D – (iv)
(2) A – (v), B – (iv), C – (iii), D – (i)
(3) A – (iii), B – (v), C – (iv), D – (i)
(4) A – (iii), B – (v), C – (vi), D – (i)

76) Division is not possible in neuron due to abscence of :-

(1) Mitochondria
(2) Centriole
(3) Nissl's granule
(4) Neurofibril

77) Ciliated epithelial cells are known to occur in :-

(1) Eustachian tube and stomach
(2) Fallopian tube and bronchioles
(3) Fallopian tube and urethra
(4) Trachea and oesophagus.

78)

How many of the following do not present in skin ?

Compound Epithelium, Dense Irregular Connective tissue, Reticular tissue, Dense regular Connective tissue.
(1) 3
(2) 2
(3) 1
(4) 4

79) Identify the correct matching pair :

(1) Microvilli – Fallopian tube
(2) Cilia – Intestine

(3) Sterocilia – Gall bladder

(4) Ependymal epithelium – Inner lining of ventricle of brain

80) In all connective tissue except ___A___ the cells secrete fibres of structural proteins called collagen or elastin.

(1) Bone

(2) Blood

(3) Areolar Tissue

(4) Cartilage

81) How many osteocyte found in lacunae of bone?

(1) 1–2

(2) 2–3

(3) 1–4

(4) Only 1

82) Most abundant connective tissue which serves as a supporting frame-work for epithelium is ?

(1) Adipose tissue

(2) Areolar tissue

(3) Dense regular CT

(4) Blood

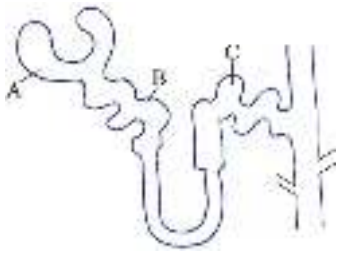
83) Which cells are also called as cart wheel cells?

(1) Plasma cells

(2) Adipose cells

(3) Mast cells

(4) Macrophages

84)  Identify type of Epithelium tissue at the label A, B, and C

	A	B	C
1	Simple cuboidal	Simple ciliated columnar	Simple Brush border cuboidal
2	Simple squamous	Simple brush bordered cuboidal	Simple cuboidal
3	Simple squamous	Simple ciliated cuboidal	Simple cuboidal
4	Simple cuboidal	Simple Brush Bordered cuboidal	Simple squamou

(1) 1

(2) 2

(3) 3
(4) 4

85) Basement membrane can not be seen in

(1) Simple cuboidal epithelium
(2) Simple columnar epithelium
(3) Simple squamous epithelium
(4) Transitional epithelium

86)

White fibrous cartilage is present in :-

(1) Larynx
(2) Articular cartilage
(3) Intervertebral disc
(4) Nasal septum

87) How many of the following are examples of simple cuboidal epithelium?
Proximal convoluted tubule, Moist surface of Buccal cavity, Urinary bladder, Stomach, Lining of follopian tube, Ducts of glands

(1) Two
(2) Three
(3) Four
(4) Five

88) Which of the following is not a property of epithelial tissues ?

(1) Cells are compactly packed with large amount of intercellular matrix
(2) Consist of free surfaces, which faces either a body fluid or the outside environment
(3) Provide covering or lining for some part of the body.
(4) Simple epithelium is composed of a single layer of cells

89)

Which of the following statements is incorrect ?

(1) Cartilage and bone both are structural tissue
(2) Blood is a fluid tissue with transport functions
(3) Areolar tissue is main reservoir of stored energy
(4) Muscle tissue, which becomes contracts in response to stimulation

90)

The fibres provide strength to connective tissue. Here 'fibres' word is use for :-

(1) White fibre
(2) Elastic fibre
(3) Reticular fibre
(4) 1 & 2 both

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	1	4	2	1	2	3	3	3	3	3	3	3	2	3	3	3	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	1	1	3	4	4	4	1	1	2	1	2	3	4	4	4	3	1	2
Q.	41	42	43	44	45															
A.	2	2	3	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	1	4	2	1	1	2	2	3	3	2	1	3	4	3	2	2	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	1	1	1	1	4	2	3	3	2	4	1	1	1	3	2	2	2	1
Q.	86	87	88	89	90															
A.	4	2	2	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	1	1	2	3	1	2	3	4	4	3	4	2	1	3	1	3	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	2	1	2	4	2	4	3	3	3	3	4	1	3	3	1	1	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	1	4	2	1	1	3	1	3	2	2	4	2	1	3	3	2	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	3	4	2	3	2	1	4	2	4	4	3	3	3	2	2	2	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	1	2	4	3	1	1	3	1										

SOLUTIONS

PHYSICS

1) $[KE] = [ML^2T^{-2}]$

K.E. = $(100\ g)\left(\frac{1cm}{10s}\right)^2$ $=1\ g\ cm^2/s^2$

$= 1$ erg

2) **Question Explain:**

The question asks to find the maximum velocity acquired by a car that accelerates from rest at a constant rate $\boldsymbol{\alpha}$, then decelerates at a constant rate β, and comes to rest after a total time $\boldsymbol{t}$.

Concept:

This problem involves the concepts of uniform acceleration and deceleration. The maximum velocity is achieved at the point where acceleration ends and deceleration starts.

Formula Used:

1. $\boldsymbol{v = u + at}$ (for velocity under constant acceleration)
2. $\boldsymbol{v^2 = u^2 + 2}$as (for velocity and displacement under constant acceleration)

Calculation:

A. During the acceleration phase, let the maximum velocity be $\boldsymbol{v_{max.}}$
B. The car accelerates for a time $\mathbf{t_1}$ so the final velocity after this phase is $\boldsymbol{v_{max} = \alpha t_1.}$
C. The car then decelerates for a time $\boldsymbol{t_2}$ and comes to rest, so $\boldsymbol{0 = v_{max} - \beta t_2}$ **, giving** $\boldsymbol{t_2 =}$ $\frac{V_{max}}{\beta}$
D. The total time $\boldsymbol{t = t_1 + t_2}$**, so** $\boldsymbol{t =}$ $\frac{V_{max}}{\alpha} + \frac{V_{max}}{\beta}$.
E. Solving for $v_{max'}$ we get:

$$V_{max} = \frac{\alpha\beta}{\alpha + \beta}t$$

moderate

3)

$T = P^a\ D^b\ S^c$

$T = [ML^{-1}\ T^{-2}]^a\ [ML^{-3}]^b\ [MT^{-2}]^c$

$M^0L^0T^1 = M^{a+b+c}\ L^{-a-3b}\ T^{-2a-2c}$

$a + b + c = 0$

$-a - 3b = 0$

$-2a - 2c = 1 \Rightarrow a + c = -\frac{1}{2}$

$b = \frac{1}{2}$, $a =$, $c = 1$

$a = \frac{1}{2}$, $b =$, $c = 1$

$a = -\frac{3}{2}$

4)

Explanation:

We need to determine the numerical values of a 10N force when expressed in a system where length unit = 10 cm, mass unit = 10 g, time unit = 0.1 sec.

Concept:

Conversion from one system to another.

Formula:

$$N_2 = N_1\left[\frac{M_1}{M_2}\right]^x\left[\frac{L_1}{L_2}\right]^y\left[\frac{T_1}{T_2}\right]^z$$

Mathematical Calculation:

$$N_2 = 10\left[\frac{1\,kg}{10\,g}\right]^1\left[\frac{1\,m}{10\,cm}\right]^1\left[\frac{1\,s}{0.1\,s}\right]^{-2}$$

$$N_2 = 10\left[\frac{1000\,g}{10\,g}\right]^1\left[\frac{100\,cm}{10\,cm}\right]^1\left[\frac{1\,s}{0.1\,s}\right]^{-2}$$

$N_2 = 10^1 \times 10^2 \times 10^1 \times 10^{-2}$

$N_2 = 10^2 = 100$ unit

Correct answer with option : 4

Estimated Time : 2-3 minutes

Level : Medium

5) Least count = $\frac{1\,mm}{10}$ = 0.1 mm;

Zero error = –(10–6) × 0.1 = – 0.4 mm

Reading = 6 + 5 × (0.1) – (–0.4) = 6.9 mm

Now $\frac{x}{10} = 6.9 \Rightarrow \boxed{x = 69}$

6)

Option (1) is correct

7) Ans. Statement I is true but Statement II is false.

8) $\frac{[x]}{[t]^4} \Rightarrow \frac{[L']}{[T']^4} = [L'T^{-4}]$

9)

$1013 = 1.013 \times 10^3$
Order of magnitude = 3

10)

If θ is angle between $\vec{P}$ and $\vec{Q}$
$\Rightarrow R_1^2 = P^2 + Q^2 + 2PQ\cos\theta$ ... (i)
then (80 – θ) will be angle between and $\vec{P}$ and $\vec{Q}$
$\Rightarrow R_2^2 = P^2 + Q^2 - 2PQ\cos\theta$ ...(ii)
(i) + (ii)
$\Rightarrow R_1^2 - R_2^2 = 2(P^2 + Q^2)$

11)

12) **Question is Asking About:**
In this question we have to choose the correct statement regarding a dimensionless quantity.

Concept:
Dimension of physical quantities

Solution/Explanation/Calculation:
A dimensionless quantity may have unit for example, Angle or solid angle are dimensionless quantities but still have units (radian and steradian).

Hence, option (3) is correct.
Conclusion : Hence, option (4) is correct

Estimated Time : 1 Minute.

Level : Easy

13)

According to law of homogeneity,
No two physical quantities having differensions can be added or subtracted.

14)

A: Question explanation
The question asks us to calculate the least count of a screw gauge.
B: Given data

A. Pitch of the screw = 0.1 cm
B. Number of divisions on the circular scale = 200

C: Concept

The least count of a screw gauge is calculated by dividing the pitch of the screw by the number of divisions on the circular scale. This is because each division on the circular scale corresponds to a fraction of the pitch.

D: Mathematical calculation

$$\frac{\text{Pitch}}{\text{Number of derivision on circular scale}}$$

A. Convert the pitch to millimeters: 0.1 cm = 1 mm
B. Calculate the least count: Least count = Pitch / Number of divisions = 1 mm / 200 = 0.005 mm

E: final answer

The least count of the screw gauge is 0.005 mm. Therefore, option 3 is the correct answer.

F : question level

Easy

15) $F \propto A^x V^y \rho^z$

$$\frac{k\ell}{ma} = \text{Dimension less} \qquad \ldots.(1)$$

16)

$$\text{Dim. of } \frac{ma}{K} = \text{Dim. of b} \ldots.(2)$$

17) $m \propto E^a v^b F^c$

a = 1, b = –2, c = 0.

18)

Option (3) is correct

19)

$$\text{Given, } \left|\vec{F}_1\right| = 12N, \ \left|\vec{F}_2\right| = 5N$$

$$\left|\vec{F}_1 + \vec{F}_2\right| = 13$$

$$\therefore (13)^2 = (12)^2 + (5)^2 + 2 \times 12 \times 5 \cos\theta$$

$$\therefore \theta = \frac{\pi}{2} = 90^\circ$$

20)

Option (2) is correct

21)

(where $\frac{dv}{dr}$ is the velocity gradient with dimensions $[t^{-1}]$).

$$\eta = \frac{F}{A\frac{dv}{dr}}$$

$$[\eta] = \frac{[MLT^{-2}]}{[L^2T^{-1}]} = [ML^{-1}T^{-1}]$$

22) $\frac{EL^2}{M^5G^2} = [1]$

 Angle

23)

$E = G^p h^q c^r$
$G = [M^{-1}L^3T^{-2}]$
$h = [ML^2T^{-1}]$
$c = [LT^{-1}]$
$E = [M^{-1}L^3T^{-2}]^p\ [ML^2T^{-1}]^q [LT^{-1}]^r$
$[ML^2T^{-2}] = [M^{-p+q}\ L^{3p+2q+r}\ T^{-2p-q-r}]$
Comparing power both side
$-p + q = 1$ ...(1)
$3p + 2q + r = 2$ ...(2)
$-2p - q - r = 2$ ...(3)
Equation (2) + (3)
$p + q = 0$
$p = q$
by (1) and (4)
$-(-q) + q = 1$
$q = 1/2$
$p = -1/2$
from equation (3)
$-2\ (-1/2) - 1/2 - r = -2$
$+1 - 1/2 - r = -2$
$r = 5/2$

24)

$a = Kt$

$$\frac{dv}{dt} = Kt$$

$$\int_u^v dv = k\int_0^t tdt$$

$$v - u = \frac{Kt^2}{2}$$

$$v = \frac{Kt^2}{2} + u$$

$$\frac{dx}{dt} = \frac{Kt^2}{2} + u$$

$$\int dx = \int \frac{Kt^2}{2} dt + \int u dt$$

$$S = ut + \frac{Kt^3}{6}$$

25) $x = a - b$

$\Rightarrow \Delta x = \Delta a + \Delta b$

$$\therefore \%\Delta x = \frac{\Delta x}{x} \times 100$$

$$= \left[\frac{\Delta a + \Delta b}{a - b}\right] \times 100$$

26)

Option (4) is correct

27) $v = \frac{D}{t} \Rightarrow v = \frac{24}{6} = 4\,m/s$

and $\frac{\Delta v}{v} = \frac{\Delta D}{D} + \frac{\Delta t}{t} \Rightarrow \Delta v = 0.1\ m/s$

$v = (4 \pm 0.1)\ m/s$

28) 30 Divisions of vernier scale coincide with 29 divisions of main scales

Therefore 1V.S.D = $\frac{29}{30}$MSD

Least count = 1MSD – 1VSD = 1MSD – $\frac{29}{30}$MSD

$= \frac{1}{30}MSD = \frac{1}{30} \times 0.5^{o} = 1\ minute.$

29) $4.338 + 4.835 \times 3.88 \div 3.0$

$= 4.338 + 4.835 \times 1.3$

$= 4.338 + 6.3$

$= 4.3 + 6.3 = 10.6$

$T = k\rho^x d^y E^z$

$T = \left[ML^{-1}T^{-2}\right]^x \left[ML^{-3}\right]^y \left[ML^2T^{-2}\right]^z$

$x + y + z = 0 \quad ...(1)$

$-x - 3y + 2z = 0 \quad ...(2)$

$-2x - 2z = 1 \quad ...(3)$

$x + z = -\frac{1}{2} \quad ...(4)$

$-y = -\frac{1}{2} \Rightarrow y = \frac{1}{2}$

by equation (2) $-x - \frac{3}{2} + 2z = 0$

$-x + 2z = \frac{3}{2} \quad ...(5)$

Adding (4) & (5)

$3z = 1, z = \frac{1}{3}$

$\Rightarrow x = -\frac{1}{2} - \frac{1}{3} = -\frac{5}{6}$

30) Ans.: A

31) (i) Length (L), mass (M) and velocity (LT^{-1}) are independent.
(ii) Pressure ($ML^{-1}T^{-2}$), density (ML^{-3}) and velocity (LT^{-1}) are dependent as

$$\begin{vmatrix} 1 & -1 & -2 \\ 1 & -3 & 0 \\ 0 & 1 & -1 \end{vmatrix} = 1(3-0) + 1(-1-0) - 2(1-0) = 0$$

(iii) Force (MLT^{-2}), velocity (LT^{-1}) and time (T) are independent as
(iv) $\text{Force} = \frac{\text{momentum}}{\text{time}}$ ⇒ force, momentum and time are dependent.

32) The mean value of refractive index,

$$\mu = \frac{1.34 + 1.38 + 1.32 + 1.36}{4} = 1.35$$

and

$$\Delta\mu = \frac{|(1.35 - 1.34)| + |(1.35 - 1.38)| + |(1.35 - 1.32)| + |(1.35 - 1.36)|}{4}$$

$= 0.02$

Thus $\frac{\Delta\mu}{\mu} \times 100 = \frac{0.02}{1.35} \times 100 = 1.48\%$

33) $\frac{2\pi\beta}{v^2}$ is input of logarithmic function

$\Rightarrow \frac{[2\pi\beta]}{[v^2]} = 1 \quad \Rightarrow \frac{[\beta]}{[v^2]} = 1$

$\Rightarrow [\beta] = [v^2] = (LT^{-1})^2 = L^2T^{-2}$

logarithmic functions are dimensionless

$$\Rightarrow [\alpha] = \frac{[F]\left[v^2\right]}{[\beta]^2} = \frac{MLT^{-2} \times \left[LT^{-1}\right]^2}{\left[L^2T^{-2}\right]^2}$$

$\Rightarrow [\alpha] = ML^{-1}T^{0}$

34) $1\ J = n_2\ u_2$

$$n_2 = 1\frac{J}{u_2} = 1\left[\frac{m_1}{m_2}\right]\left[\frac{L_1}{L_2}\right]^2\left[\frac{T_1}{T_2}\right]^{-2}$$

$$n_2 = 1\left[\frac{kg}{100g}\right]\left[\frac{m}{10\ cm}\right]^2\left[\frac{sec}{0.5\ sec}\right]^{-2}$$

$$= 1\left[\frac{1000\ gm}{100\ gm}\right]\left[\frac{100\ cm}{10\ cm}\right]^2[0.25] =$$

$n_2 = 250$

1 Joule = $(250)u_2$

$$u_2 = \frac{1}{250}J = 0.004\ J$$

35) $$\frac{1}{Req} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{dReq}{R^2eq} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2}$$

$$\Rightarrow \frac{dReq}{Req} = R\,eq\left\{\frac{dR_1}{R_1} \times \frac{1}{R_1} + \frac{dR_2}{R_2} \times \frac{1}{R_2}\right\}$$

$$\%\ error\ = 2\left\{\frac{1}{3} + \frac{2}{6}\right\}$$

$$= \frac{4}{3}\% = 1.33\%$$

36) $$Density\ (\rho) = \frac{mass\ (m)}{Volume\ (V)} \qquad \Delta m = 0.05kg,\ m = 5\ kg \qquad \Delta V = 0.05m^3, V = 1m^3$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$$

$$= \left(\frac{0.05}{5}\right)100\% + \left(\frac{0.05}{1}\right)100\%$$

$= 1\ \% + 5\% = 6\ \%$

37) $$\left.\begin{array}{ll} [a] = \alpha\beta^{-2} \Rightarrow LT^{-2} = \alpha\beta^{-2} & \\ [v] = \alpha\beta^{-1} \Rightarrow LT^{-1} = \alpha\beta^{-1} & \Rightarrow\ [\beta] = T \\ [\ell] = \alpha\gamma \Rightarrow L = \alpha\gamma & \end{array}\right] \begin{array}{l} [\alpha] = L \\ \\ [\gamma] = 1 \end{array}$$

Dimension of coefficient of friction = 1

Only option (4) is dimensionless.

38) $P = \vec{F} \cdot \vec{v}$

$F \to \frac{F}{2}$; $v \to \frac{v}{2}$

Then $P \to \frac{P}{4}$

39)

L.C. = $\frac{\frac{1}{2}mm}{50} = \frac{1}{100}$ mm

Diameter = 3mm + 35 (.01 mm) + 0.03 mm = 3.38 mm

40) Measured value = MSR + LC × VSR = 12 mm + 0.2 mm × 3
= 12.6 mm = 1.26 cm

41) Traiveling zero are not significant figure.

42) Here, S = (13.8 ± 0.2) cm; t = (4.0 ± 0.3)s

$\therefore \quad V = \frac{13.8}{4.0} = 3.45\,ms^{-1}$

Also, $\frac{\Delta V}{V} = \pm\left(\frac{\Delta S}{S} + \frac{\Delta t}{t}\right)$

$= \pm\left(\frac{0.2}{13.8} + \frac{0.3}{4.0}\right) = \pm 0.0895$

$\Delta V = \pm 0.0895 \times 3.45 = \pm 0.3$

(rounding off to one place of decimal)

$V = (3.45 \pm 0.3)\ m\ s^{-1}$.

43) $L.C. = \left(\frac{b-a}{b}\right) M = \frac{[n-(n-1)]}{n}x$

$= \frac{1}{n}x = \frac{x}{n}cm$

44) Assertion & Reason both are correct
Theory

The correct option is **A.** M^0LT^{-1} and T^{-1}

Given,

$x = \frac{V_0}{a}(1 - e^{-at})$

'at' should be dimensionless $\Rightarrow$ Dim (a) $= \frac{1}{\text{Dim}(T)} = T^{-1}$

x is position $\Rightarrow$ Dim (x) $= M^0LT^0 =$ Dim $(\frac{V_0}{a})$

45) $\therefore$ Dim (V_0) = Dim (x) × Dim (a) $= M^0LT^{-1}$.

$2KClO_3 \rightarrow 2KCl + 3O_2$
$12.25\ g$

46) $= \frac{12.25}{122.5} = 0.1\ mol$

$no_2 = 0.15\ mol$
$Vo_2 = 0.15 \times 22.4 = 3.36\ L$

47) We know that in the formation of carbon monoxide (CO), 12 parts of carbon by weight (equal to its atomic weight) combine with 16 parts of oxygen by weight (equal to its atomic weight).
Similarly, in carbon dioxide (CO_2) 12 parts of carbon by weight (equal to its atomic weight) combine with 32 parts of oxygen (O_2) by weight (2 Ã— atomic weight of oxygen).
Thus different weights of oxygen (16 and 32) combine with a constant weight of carbon (12) in the ratio 16 : 32 or 1 : 2 to form carbon monoxide and carbon dioxide in a simple integral ratio.
Thus CO and CO_2 illustrates the law of multiple proportion.

48) Given : Atomic weight of element (M) = 15 and vapour density of its chloride (V.D.) = 60
We know that molecular weight of compound = 2 Ã— V.D. = 2 Ã— 60 = 120
Formula of compound is MCl_x
M + 35.5 x = 120
15 + 35.5 x = 120
$x = \frac{105}{35.5} = 2.9 \simeq 3$
valency = 3

$1m^3 = 1000\ L$

49) $\text{number of moles} = \frac{1000}{22.4} = 44.6\ \text{mole}$

$18\ g\ \text{of water} = 6 \times 10^{23}\ \text{molecule}$
$1\ g\ \text{water} = 1\ ml\ \text{water}$

50) $= \frac{6}{18} \times 10^{23}\ \text{molecule} = 20\ \text{drops}$

$\therefore\ 1\ \text{drops consist} = \frac{6}{18} \times \frac{1}{20} \times 10^{23}$

51) Mass of 1 L of N_2 at NTP = Mass of compound
$\frac{1}{22.4} \times 28 = \frac{1}{22.4} \times M$
M = 28

52) $2CHI_3 + 6Ag \rightarrow C_2H_2 + 6AgI$

2 mol $CHI_3 \rightarrow$1mol C_2H_2

0.01 mol $CHI_3 \rightarrow \frac{0.01}{2}$ = 0.005 mol

=0.05 × 22400 = 112 ml

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\frac{112}{273} = \frac{V_2}{546}$

224 ml $= V_2$

53)

mol. wt = 2 Ã— V.D.

= 2 Ã— 70 = 140

(12 + 16) Ã— x = 140

28x = 140

x = 5

54)

$(gea)_M$=(gea)H_2

$\frac{12}{24} = \frac{V}{22.4}$

$\frac{1}{2} = \frac{V}{11.2}$

V = 5.6 L

55)

$H_2 + Cl_2 \rightarrow 2HCl$ Cl_2 is L.R.

12L 11.2 L

0.8 – 22.4

56)

1 mol $\rightarrow$ 60 g CO_3^{-2}

0.016 mol $\rightarrow$ 60 × 0.16

=0.96 g CO_3^{-2}

0.4 mol P atom

1 mol $Na_3(Po_4)_2$ required 2 mol P atoms

57) No. of moles of $Na_3(PO_4)_2$ formed $= \frac{0.4}{2} = 0.2$ mol

$= 0.2 \times 6 \times 10^{23}$ $Na_3(PO_4)_2$ molecule

$= 1.2 \times 10^{23}$

58)

1 mol $\rightarrow$ 3 N_A
0.01 mol $\rightarrow$ 3 N_AO atom

59)

			Moles	Ratio
Na	28.5	23	1.24	1
S	40	32	1.25	1
O	31.5	16	1.96	1.5

Na : S : O
1 : 1 : $\frac{3}{2}$

$Na_2S_2O_3$

60) Given : Weight of sodium bicarbonate ($NaHCO_3$) = 6.3 g ; Weight of hydrochloric acid (HCl) We know that

$$\underset{\text{(Sodium bicarbonate)}}{NaHCO_3} + \underset{\text{(Hydrochloric acid)}}{HCl} \longrightarrow x + \underset{\text{(Carbon dioxide)}}{CO_2}$$

Thus weight of rectants = Weight of $NaHCO_3$ + Weight of HCl = 6.3 + 10.0 = 16.3 g
And weight of products = Weight of x + Weight of CO_2 = 12.0 + CO_2.
We also know from the law of conservation of mass that weight of rectants = Weight of product.
Therefore 16.3 = 12.0 + CO_2 or CO_2 = 16.3 â€“ 12.0 = 4.3
Therefore weight of CO_2 released = 4.3 g.

61)

$$S + O_2 \xrightarrow{50\%\ \text{yield}} SO_2$$
a a/2

$$NO_2 + SO_2 + H_2O \xrightarrow[\text{yield}]{80\%} H_2SO_4 + NO$$
a/2 0.4a

0.4a × 98 = 98 = 1960
$a = \frac{1960}{98 \times 0.4}$
a = 50
Mass = a × 32
g 'S' = 50 × 32 = 1600 g

$$\frac{N_{CO_2}}{N_{N_2O}} = \frac{n_{CO_2} \times N_A}{n_{N_2O} \times N_A}$$

62) $$= \frac{wt(CO_2) \times N_A}{Mw(CO_2)} \times \frac{Mw(N_2O)}{wt(N_2O) \times N_A}$$

$$\frac{2 \times N_A}{44} \times \frac{44}{5 \times N_A} = 2 : 5$$

Let at wt of metal = x

oxide :- I

M O

63) $$\frac{50}{x} : \frac{50}{16}$$

formula = MO

Hence x = 16

In oxide II : M O

formula of II oxide M_2O_3 $\frac{40}{16}$ $\frac{60}{16}$

2 : 3

64) $$\frac{\left(\frac{5.72\ gm}{x}\right)}{\left(\frac{10 - 5.72 = 4.28\,gm}{16}\right)} = \frac{4}{6}$$

$\Rightarrow X = 32$ amu

65)

	Mass	Molar Mass	Moles	Ratio
P	254	127	2	P : Q
Q	80	16	5	2 :5

P_2Q_5

66) Assume total mass = 200 gm

$\therefore$ Mass of $CaCO_3$ = 100 gm

Then loss in mass = 44 gm

$\therefore$ Percentage loss = 22%

67)

$$\text{Volume(ml)} = \text{Mass (g)}$$
$$22400 = 44$$
$$1120 = \frac{1120}{22400} \times 44 = 2.2 \text{ g}$$

68) $W_{H_2O} = 175g$

$\frac{175}{18} N_A$ = No. of H_2O molecules

For ethanol

$\frac{W}{16} = \frac{175}{18}$

Wt = 447.2

$V = \frac{W}{D} = \frac{447.2}{0.8}$

= 559 mL

69) Mass of 1 L of $N_2 = \frac{28}{22.4} = 1.25$ g

70)

	Mass	Molar mass	Moles	Spefification
C	38.8	12	3.23	1
H	16	1	16	5
N	45.2	14	3.23	1

CH_5N

71)

4g He $\rightarrow 6 \times 10^{23}$ atom

1g He $\rightarrow \frac{6 \times 10^{23}}{4} = 1.5 \times 10^{23}$atom

72)

Fe_2O_3 FeO $O_1 : O_2$

$Fe \frac{O_3}{2}$ $\frac{3}{2} : 1$

$3 : 2$

73) Mass of Metal Chloride $MCl_x = M + x \times 35.5$

$= 2 \times V.D$

$= 79$

$M\omega = E\omega \times x$

$MCl_x = 3.82 \times x + x \times 35.5 = 79$

$x = 2$

Atomic Weight $= 3.82 \times 2 = 7.64$

74) Let M is molecular weight

moles of Br $= \frac{100}{M} \times 3$

wt. of Br $= \frac{100}{M} \times 3 \times 80 = 10.46$

$M = 2295.45 = 240 + 75 + 44n$

$n = 45$

75) Mass of 0.05 Moles of B_2A_3 = 9 g

Mass of 1 mole of $B_2A_3 = \frac{9}{0.05} = 180$ g

Mass of 0.10 moles of B_2A = 10 g

Mass of 1 mole of $B_2A = \frac{10}{0.10} = 100$ g

$2B + 3A = 180$ g ....(i)

$2B + A = 100$ g ....(ii)

$A = 40$ g

$B = 30$ g

76) Correct answer is (B)

77) **Question Explanation:** The assertion compares the number of molecules in 88g of CO_2 and 72g of H_2O starting that CO_2 has more molecules. The reason given is that both have 4 moles of molecules. We need to verify both the assertion and the reason.

Concept:

The number of moles is determined by dividing the mass of the substances by its molar mass. The number of molecules in a sample is directly proportional to the number of moles.

Solution:

- The molar mass of CO_2 is 44g/mol, so 88g of CO_2 contains.
- $\frac{88}{44} = 2$ 2 moles of CO_2 molecules.
- The molar mass of H_2O is 18g/mol, so 72g of H_2O contains:

$\frac{72}{18} = 4$ 4 moles of H_2O molecules.

Since CO_2 has 2 moles of molecules and H_2O has 4 moles of molecules, the assertion is incorrect.

Additionally the reason starting that both have 4 moles of molecules is also incorrect.

Final Answer

Both the reason and the reason are incorrect

Correct option: (4)

78)

Specific gravity is dimensionless as it is relative term dimensions of density is cancelled by dimensions of other density.

79)

3.2 g 'O' in $\rightarrow$ l00 g compound

16 g 'O' in $\rightarrow = \frac{100 \times 16}{3.2}$ 500 g

Minimum molecular mass is the mass of compound which have at least 1 atom.

80)

CH_4 : O_2 : SO_2

$\frac{W}{16} \quad \frac{W}{32} \quad \frac{W}{64}$

$\frac{1}{16} : \frac{1}{32} : \frac{1}{64}$

4 : 2 : 1

81)

Avg. atomic mass is always between highest and lowest isotope more the abundance of particular isotope avg. atomic mass is near to that.

82)

1 gm molecule = 1 mole

1 mole = 5 N_A atoms

83)

Option (2) is correct

84)

$$\text{Equivalent Mass} = \frac{\text{Molar Mass}}{\text{no. of } H^+/OH^- \text{ replaced in a reaction}}$$

$\rightarrow H_3PO_4$ can give $3H^+$ hence it is tribasic acid

85)

Option (1) is correct

86)

$1e^-$ mass = $9.1 \times 10^{-31} \times 10^3$ g
6×10^{23} e^- mass = $9.1 \times 10^{-31} \times 6 \times 10^{23} \times 10^3$g molar mass

$$n = \frac{1000}{9.1 \times 10^{-31} \times 6 \times 10^{23} \times 10^3}$$

$$= \frac{10^8}{9.1 \times 6}$$

87)

$V_{Initial}$ = 20 + 120 = 140 ml
V_{final} after Ist contraction = 140 – 60 = 80 ml
$V_{CO_2} = V_{KOH}$ absorbed = 60 ml
V'_{Final} after 2nd contraction = 80 – 60 = 20 ml
This is 20 ml is vol. of unreacted O_2
so V_{O_2} (reacted) = 100 ml

$$C_xHg_{(g)} + \left(x + \frac{y}{4}\right) O_2 \rightarrow XCO_{2(g)} + \frac{y}{2}H_2O_{(l)}$$

20ml $\overset{20\left[x+\frac{y}{4}\right]}{\quad}$ $\overset{20x}{\quad}$
VCO_2 = 20 x = 60 = x = 3
$VCO_2 = \overset{20\left[x+\frac{y}{4}\right]}{\quad}$ = 100 = y = 8
C_3H_8

88) $E_{X_2Y} = \frac{X}{1} + \frac{Y}{2} = 38,\ 2X + Y = 76$

$E_{X_2Y_3} = \frac{X}{3} + \frac{Y}{2} = 18,\ 2X + 3Y = 108$

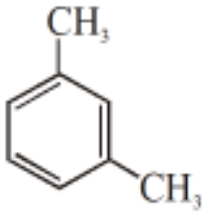
On solving,
X = 30, Y = 16

89)

$$3Fe_2O_3(s) \rightarrow 2Fe_3O_4 + \frac{1}{2} O_2$$

480 g Fe_2O_3 provide 16 g O_2.
For loss of 0.04 g $O_2 \rightarrow 0.04 \times \frac{480}{16}$ = 1.2 g
Fe_2O_3% by mass of

$$SiO_2 = \frac{0.8}{2.0} \times 100 = 40\%$$



90) $\text{mass} = 2.12\ \text{gm} + \frac{21}{2} O_2 \rightarrow 8CO_2 + 5H_2O$

ratio of $\frac{\text{O}-\text{atom}}{\text{H}-\text{atom}} = \frac{(16+5)}{10} = \frac{21}{10}$

BIOLOGY

91) NCERT-XI Pg. # 131

92) NCERT Pg # 132

93)

Lipids are arranged within the cell membrane with the hydrophobic tails towards the inner part because.
This ensures that non-polar tail of saturated hydrocarbons is protected from the aqueous environment.

94) Active transport occurs from low concentration to high concentration.

95)

Plasmodesmata are cytoplasmic connection between two cells.

96)

Cell wall in plant is non-living and Mainly made up of polysaccharide

97)

Ribophorin is glycoprotein present on cisternae of ER

98)

(A) Animal cells contain non-membrane bound organelle called centriole which help in cell division.
(B) Lipids are arranged within the plasma membrane with the polar head towards the outer sides and the hydrophobic tails towards the inner side.

99)

The stroma of chloroplast contains:-
Small, double stranded DNA molecule

100) NCERT Pg.#135

101)

NCERT (XIth) Pg. # 133, 8.5.3.1, 8.5.3.2

102)

Mitochondria are semi-autonomous.

103) NCERT Pg. # 132

104) NCERT (XI) Pg. # 95, fig.-8.5

105)

Double membrane and 70S ribosomes are found in both plastid and mitochondria

106)

Carbohydrates < Proteins > Lipids

107)

Porins are present in outer membrane of plastid and mitochondria

108)

Middle lamella is present outside the primary wall

109) Chlorophyll pigments are present in thylakoid and enzymes for the synthesis of carbohydrates are found in stroma

110)

The correct answer is:
4. All of the above
Explanation:
The fluid nature of the cell membrane refers to the ability of the lipid bilayer to allow movement of lipids and proteins within the membrane. This flexibility is crucial for several

cellular processes, including:

1. Cell growth and division:

A. The fluidity of the membrane allows it to expand and reorganize during cell growth and division, enabling ·the formation of new membranes during cell division.

2. Formation of intercellular junctions:

A. The fluidity of the membrane allows cells to interact and form junctions like tight junctions, gap junctions, or desmosomes, which are essential for cell-to-cell communication and adhesion.

3. Secretion and endocytosis:

A. Membrane fluidity facilitates processes like vesicle formation (for secretion) and vesicle fusion with the membrane (for endocytosis), allowing the cell to take in or release substances.

Thus, the fluid nature of the membrane is key to all of these processes.

111)

XIth NCERT OLD, Pg. 132 - 133

112)

Tertiary cell wall is present in some plant cells

113)

NCERT STATEMENT

114) NCERT Pg. # 131

115) NCERT Pg. # 132

116)

NCERT STATEMENT

117)

NCERT STATEMENT

118)

NCERT STATEMENT

119)

NCERT STATEMENT

120)

NCERT STATEMENT

121)

NCERT STATEMENT

122)

NCERT STATEMENT

123)

Module pg - 182

124)

NCERT STATEMENT

125)

Mitochondria :-
(b) are typically sausage-shaped or cylindrical
(c) are double membrane bound structures
(d) have two aqueous compartments

126)

NCERT STATEMENT

127)

NCERT STATEMENT

128) NCERT Pg#134

129) NCERT XI Pg # 131

130) NCERT Pg. # 133

131) NCERT XI Pg.# 135

132) NCERT XI Pg.# 136

133)

NCERT Pg. # 136

134)

NCERT STATEMENT

135)

NCERT STATEMENT

136)

The correct statement is
(a) It is characterized by a single large droplet of fat in the cytoplasm of its cells.
(d) Cytoplasm is pushed to periphery
Explanation: This describes the characteristic feature of white adipose cells, which are the primary fat cells in the body. These cells store large amounts of fat in a single, large droplet within their cytoplasm, pushing the nucleus and other organelles to the periphery of the cell. This is a consequence of the large fat droplet in white adipose cells, not a separate statement. The single large lipid droplet occupies most of the cell's volume, pushing the cytoplasm and organelles to the edges.

137)

Old NCERT Page # 104

138)

Adjacent epithelial cells are held together by specialized structures called cell junctions. These junctions include desmosomes, tight junctions, and adherens junctions, which play different roles in maintaining tissue integrity and function.

139)

Option (1) is correct

140) NCERT XI, Pg. # 102, IInd Para

141)

Old NCERT Page # 103

142)

Old NCERT Page # 101

143)

Old NCERT Page # 102

144)

Old NCERT Page # 104

145)

Old NCERT Page # 103

146)

Old NCERT Page # 102

147) NCERT Pg # 103

148) NCERT Pg # 102-103

149) NCERT-XII, Pg. # 103

150)

NCERT Pg # 104, 105

151) NCERT-XII, Pg. # 103

152) NCERT-Page No. # 104

153) **Correct Matching is**

List-I (Epithelial tissue)		**List-II (Location)**	
1.	Cuboidal Epithelium	**V**	Lining of pancreatic duct
2.	Ciliated Epithelium	**IV**	Inner lining of Fallopian tubes
3.	Columnar Epithelium	**III**	Inner surface of gall bladder
4.	Squamous Epithelium	**II**	Inner lining of blood vessels

5.	Keratinized Squamous	**I**	Epidermis of skin

Correct Option: (A)-(V), (B)-(IV), (C)-(III), (D)-(II), (E)-(I)
Thus, the correct answer is Option 3.

154)

Generated by Allie

Problem Statement: The question is asking which type of muscle has a special structure called intercalated discs. It provides four options where muscles are found: Brain, Spinal cord, Stomach, and Heart, and we need to identify the correct one.

Underlying Concept: Intercalated discs are unique structural features found in cardiac muscle cells. These discs are specialized connections that link adjacent cardiac muscle cells, allowing synchronized contraction of the heart. They are not present in muscles of other organs like the brain, spinal cord, or stomach.

Tips and Tricks: Remember: intercalated discs are unique to cardiac muscle, which is found only in the heart. So the keyword 'heart' should lead you to the correct answer.

Common Mistakes: Confusing muscles in other organs like the stomach (which has smooth muscle) or assuming muscles in the brain or spinal cord have intercalated discs.

Why Other Options Are Incorrect?: Brain and spinal cord contain nervous tissue, not muscle tissue. Stomach contains smooth muscle, which lacks intercalated discs. Only the heart has cardiac muscle with intercalated discs.

155)

Old NCERT Page # 105

156)

Old NCERT Page # 105

157)

Old NCERT Page # 105

158)

Old NCERT Page # 105

159)

NCERT XI, Page # 232

160)

Old NCERT Page # 105

161)

Old NCERT Page # 105

162)

NCERT-XI, Pg. # 231

163)

The correct answer is: **3. Both (1) and (2)**
Schwann cells are glial cells found in the peripheral nervous system. Their primary function is to provide support and insulation to nerve fibers.
1. **Myelinated nerve fibers**: Schwann cells wrap around the axons to form the myelin sheath, which enhances the speed of nerve impulse conduction.
2. **Unmyelinated nerve fibers**: Even in unmyelinated fibers, Schwann cells provide support by surrounding multiple axons without forming a myelin sheath.
Thus, Schwann cells are found on **both myelinated and unmyelinated nerve fibers**, making **option 3** correct.

164)

Old NCERT Page # 105

165)

Old NCERT Page # 105

166)

Old NCERT Page # 105

167) NCERT (XIth) Pg. # 101

168)

Old NCERT Page # 102

169)

Old NCERT Page # 101

170) NCERT Page No. # 103 (E/H)

171) NCERT Pg. # 104

172)

Old NCERT Page # 103

173)

Old NCERT Page # 103

174)

Old NCERT Page # 101

175)

Old NCERT Page # 101

176)

Old NCERT Page # 104

177)

Old NCERT Page # 101

178)

Old NCERT Page # 101

179)

NCERT XI, pg. # 104

180)

NCERT XI, pg. # 103